Nederlandse norm

NEN-EN 689+C1

(en)

Blootstelling op de werkplek - Meting van de inhalatieblootstelling aan chemische stoffen - Strategie om te voldoen aan de arbeidshygienische blootstellingsgrenswaarden

Workplace exposure - Measurement of exposure by inhalation to chemical agents - Strategy for testing compliance with occupational exposure limit values

Vervangt NEN-EN 689:2018

ICS 13.040.30

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EUROPEAN STANDARD

NORME EUROPÉENNE

EUROPÄISCHE NORM

EN 689:2018+AC

April 2019

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Supersedes EN 689:2018

English Version

Workplace exposure - Measurement of exposure by inhalation to chemical agents - Strategy for testing compliance with occupational exposure limit values

Exposition sur les lieux de travail - Mesurage de l'exposition par inhalation d'agents chimiques - Stratégie pour vérifier la conformité à des valeurs limites d'exposition professionnelle

Exposition am Arbeitsplatz - Messung der Exposition durch Einatmung chemischer Arbeitsstoffe - Strategie zur Überprüfung der Einhaltung von Arbeitsplatzgrenzwerten

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European foreword

This document (EN 689:2018+AC:2019) has been prepared by Technical Committee CEN/TC 137 “Assessment of workplace exposure to chemical and biological agents”, the secretariat of which is held by DIN.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by November 2018, and conflicting national standards shall be withdrawn at the latest by November 2018.

This document includes Corrigendum 1 issued by CEN on 10 April 2019.

This document includes the corrigendum 1 which modifies abbreviations, the content in several Annexes, and modifies the bibliography and references in the text.

This document supersedes [AC⟩ EN 689:2018 ⟨AC].

The start and finish of text introduced or altered by corrigendum is indicated in the text by the tags [AC⟩ ⟨AC].

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Introduction

This European Standard deals with the measurement of exposure to chemical agents in workplace atmospheres, and in particular with measurement strategy for comparing workers' exposure by inhalation with occupational exposure limit values (OELVs). Other parts of management of exposure of workers are not dealt with in detail in this European Standard.

Within this European Standard, compliance means that workers' time weighted average ▌AC⟩ occupational ⟨AC▐ exposure is below an OELV with a corresponding reference period. OELVs include legal values and other numerical criteria (see Annex B).

Representative measurement of ▌AC⟩ occupational ⟨AC▐ exposure to chemical agents is difficult, because of the variability of exposure. Processes and products affecting exposure are numerous. Different workplace conditions can correspond to different generation rates, can involve a variety of chemical agents and can therefore present specific exposure conditions. Exposure can be affected by the distance of the exposed workers from emission sources; and parameters such as emission intensity, ventilation, climatic conditions, seasonal variations and the controls applied can also have a very marked influence. The spatial and temporal variabilities of exposure conditions are further enhanced by workers' practices and activities themselves.

The sampling equipment introduces its own limitations, and the analytical steps add further difficulties and uncertainties.

This European Standard is applicable for measuring procedures that fulfil the requirements of EN 482. If a measuring procedure does not fulfil these requirements some parts of the procedure described in this European Standard cannot be applied.

To assess the exposure of workers to chemicals and to state with certainty that it does not exceed the OELVs (short-term or long-term) would require measurement of the exposure of every worker for every working day. Unfortunately while this approach is possible for some agents such as ionizing radiation, it is not feasible or practical for many chemical agents due to limitations of the measurement techniques and costs.

The strategy described in this European Standard gives a procedure for the employer or other stakeholders to overcome the problem of variability and to use a relatively small number of measurements to demonstrate with a high degree of confidence that workers are unlikely to be exposed to concentrations exceeding the OELVs. To reduce the number of exposure measurements, and therefore the cost of assessment, personal air samples are collected among workers within similar exposure groups (SEGs). A single measurement or even several measurements below the limit value can be insufficient to reliably demonstrate compliance without using a statistical test like the one proposed in this European Standard.

Respiratory protective equipment (RPE) is used to reduce the amount of the chemical agent that is inhaled by the worker. However, this European Standard does not take into account the use and effectiveness of RPE in testing compliance with the OELV.

Before any measurements are performed, it is essential for an appraiser to conduct a basic characterization in order to collect relevant information on workplace factors, and the available information on exposure in the workplace concerned. This includes information on variation of exposure with time of day and season of the year, so that the measurement is representative.

If the basic characterization shows that exposure is probably higher than the OELV, then it is recommended to reduce exposure by risk management measures (RMM) before measurements are planned for compliance testing.

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1 Scope

This European Standard specifies a strategy to perform representative measurements of exposure by inhalation to chemical agents in order to demonstrate the compliance with occupational exposure limit values (OELVs).

This European Standard is not applicable to OELVs with reference periods less than 15 min.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

EN 482, *Workplace atmospheres — General requirements for the performance of procedures for the measurement of chemical agents*

EN 1540, *Workplace exposure - Terminology*

3 Terms and definitions and abbreviations

3.1 Terms and definitions

For the purposes of this document, the terms and definitions given in EN 1540 and the following apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at http://www.electropedia.org/
- ISO Online browsing platform: available at http://www.iso.org/obp

3.1.1
appraiser
person who is sufficiently trained and experienced in occupational hygiene principles, working and measurement techniques, to conduct the part of the assessment they are performing according to the state of the art

Note 1 to entry: The appraiser may be supported by a team of qualified persons.

3.1.2
exposure profile
description of the exposure variations to a chemical agent in relation to the definable series of activities from the periods under consideration

Note 1 to entry: See Annex D.

3.1.3
similar exposure group
SEG
group of workers having the same general exposure profile for the chemical agent(s) being studied because of the similarity and frequency of the tasks performed, the materials and processes with which they work, and the similarity of the way they perform the tasks

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3.2 Abbreviations

AM	arithmetic mean
CAS	chemical abstracts service
DMEL	derived minimum effect level
DNEL	derived no-effect level
EC number	European community number
GSD	geometric standard deviation
GM	geometric mean
[AC⟩ h ⟨AC]	hour
LOQ	limit of quantification
OELV	occupational exposure limit value
RPE	respiratory protective equipment
RMM	risk management measures
SD	standard deviation
SEG	similar exposure group
WS	work shift

4 General

[AC⟩ The assessment of occupational exposure to chemical agents based on air samples requires different procedures which shall be conducted by an appraiser. ⟨AC]

[AC⟩ It starts with an initial [AC⟩ occupational ⟨AC] exposure assessment (described in Clauses 5 and 6) followed by periodic reassessments (see Clause 7) ⟨AC]. The initial [AC⟩ occupational ⟨AC] exposure assessment comprises different phases:

— basic characterization of the workplaces;

— constitution of similar exposure groups;

— selection of a suitable measuring procedure;

— performing exposure measurements;

— validation of exposure measurement results and SEGs;

— comparison of results with OELVs;

— reporting of results.

The sampling strategy comprises the constitution of SEGs and the definition of measuring procedures. If the initial [AC⟩ occupational ⟨AC] exposure assessment indicates non-compliance with OELVs for some SEGs, control measures shall be implemented and a new initial [AC⟩ occupational ⟨AC] exposure assessment shall be conducted. Following the initial [AC⟩ occupational ⟨AC] exposure assessment, periodic reassessments shall be conducted. When significant changes occur at the workplace that may affect

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exposure conditions, the appraiser shall decide whether a new initial ⟦AC⟧ occupational ⟨AC⟧ exposure assessment shall be conducted.

Figure 1 gives a schematic overview of the procedures described in this European Standard.

⟦AC⟧

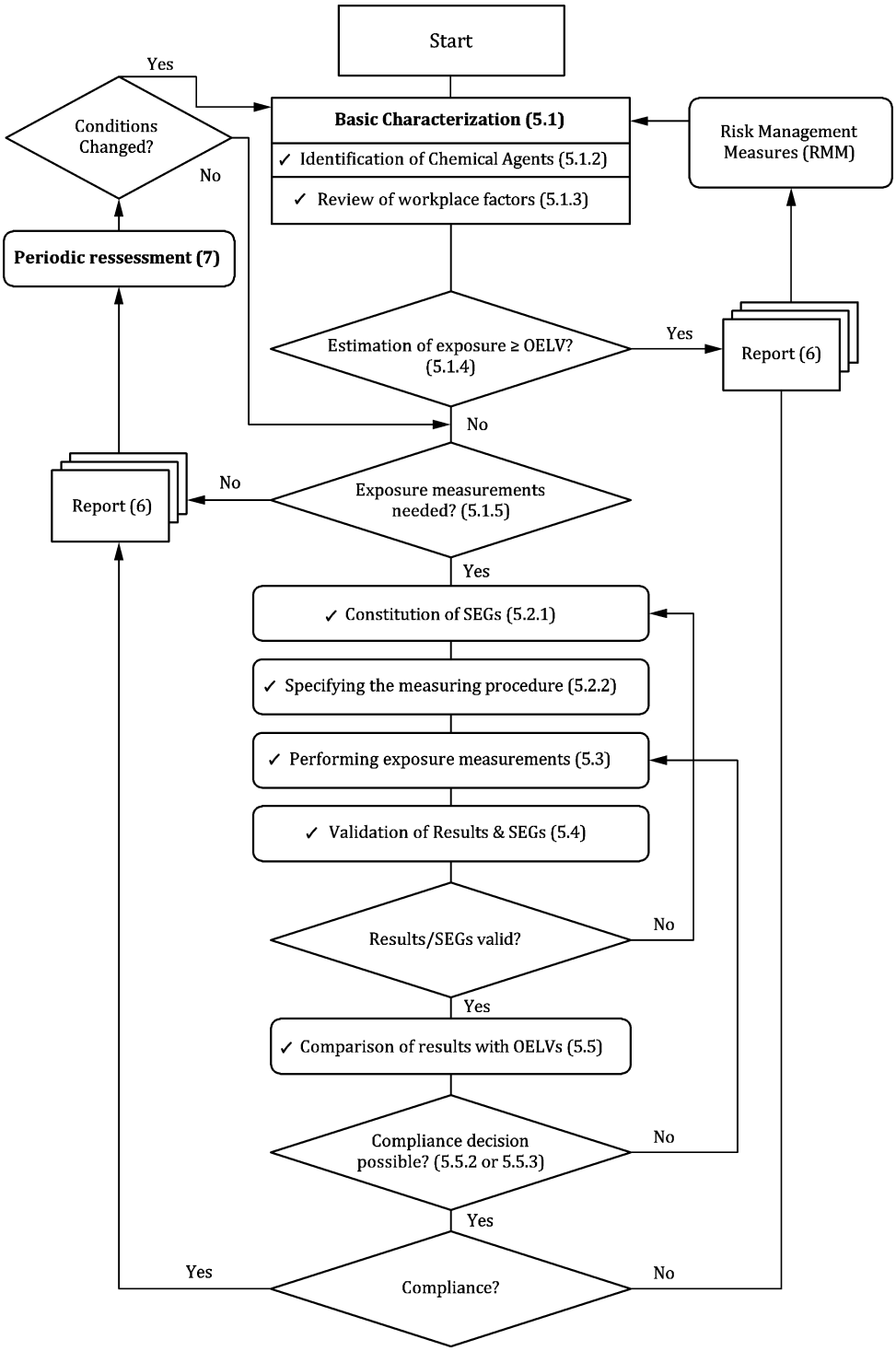

Figure 1 —Schematic overview of the occupational exposure assessment procedures ⟨AC⟧

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5 Occupational exposure assessment

5.1 Basic characterization

5.1.1 General

Prior to exposure measurements, the workplace and the related exposure profile under consideration shall be considered.

The basic characterization of exposure determinants is in three steps:

— identification of chemical agents and other information required;

— review of workplace factors;

— estimation of exposure.

All the information collected during the basic characterization shall be used to:

— decide whether measurements are necessary or not (see 5.1.4);

— constitute the different SEGs.

Use of exposure monitoring devices able to give an instant feedback to the user, either as a single or a continuous reading (so called direct-reading instruments), can give valuable information about approximate exposure level, the exposure pattern, and spatial, time and worker variability of the exposure.

5.1.2 Identification of chemical agents and other information required

The preparation of a list of all chemical agents in the workplace and the relevant information concerned is an essential first step to the identification of the potential for hazardous exposure. The safety data sheets and other available information are useful to establish the list. The list shall include any of the following:

— raw materials, primary products, impurities, intermediates, final products, reaction and process products and by-products, etc;

— the individual chemical agents, identified with chemical registration numbers (e.g. CAS#, EC#);

— hazardous properties, classification and labelling;

— appropriate OELVs depending on the goal of the assessment (see Annex B);

— whether dermal and oral exposure of the chemical agents is relevant (this European Standard only deals with measuring inhalation exposure, but knowledge of other routes can be important in the management of exposure);

— additional information such as amount used, vapour pressure, temperature, saturation concentration, dustiness, etc.

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5.1.3 Review of workplace factors

The work processes and procedures shall be evaluated to gauge the exposure and the exposure profile to chemical agents by a detailed review of workplace factors, such as:

— work organization: activity, task, job title, work shift system, job functions, etc.;

— processes and techniques (type of processes, temperature, pressure, etc.);

— workplace layout and configuration, including confined spaces, open air, etc.;

— safety precautions and procedures (restricted area, etc.);

— cleanliness and tidiness of workplace;

— ventilation installations and other forms of engineering control and any information on their performance;

— emission sources and locations of high concentrations ;

— periods, frequencies and durations of exposure taking into account variation of exposure with time of day and season of the year;

— work load;

— worker behaviour, or

— activity or production rate indicators.

5.1.4 Estimation of exposure

The appraiser shall collect available information to allow reliable estimates of the exposure of the workers. Annex A gives guidance on this.

Possible sources of information include:

— earlier measurement results in this workplace, including from direct-reading instruments;

— measurement results from comparable installations or work processes (databases, literature, etc.);

— calculations based on relevant quantitative information;

— exposure modelling.

A decision that an exposure is much lower than the OELV should be made, for example in the following cases:

— the release capability is low because of the working conditions and the substance properties (e.g. low vapour pressure, high boiling point with low processing temperature, low dust-forming behaviour);

— the operational conditions cannot give rise to the formation of aerosols;

— only small quantities are used;

— only low emissions are possible, for example because of small source surfaces or short activity duration (<15 min).

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The estimates shall be used to make a decision in 5.1.5.

The estimates shall also be used to verify whether the sampling and analysis techniques that are proposed for the measurements are adequate.

5.1.5 Decisions

The results of basic characterization lead to one of the following decisions:

— the exposure is higher than the OELV (non-compliance); then the appraiser shall report this (see Clause 6) and advise on a programme to reduce exposures, using RMM, before making measurements to test compliance;

— the exposure is well below the OELV (compliance); then the appraiser shall decide if measurements are necessary or not. If measurements are unnecessary, the appraiser shall report this (see Clause 6) and advise on a reassessment (see Clause 7);

— the available information on exposure is insufficient to decide on compliance with the OELV; then the appraiser shall continue to develop a sampling plan (see 5.2).

5.2 Sampling strategy

5.2.1 Constitution of Similar Exposure Groups (SEGs)

In practice it is usually not possible to measure the exposure of each worker during each working day. In order to obtain quantitative data from exposure measurements that allows assessment of compliance with OELVs, an effective approach shall be taken that allows the most efficient use of resources.

This approach, based on the observation of working conditions, permits measurement of exposure of a small number of workers belonging to a SEG for comparison with OELVs. Where exposure measurements on some workers of the SEG indicate that the OELVs are met (compliance), then it is considered that this is so for all workers in the SEG.

The SEG shall be constituted using information on the profile of exposure and duration of the tasks performed in the working shifts throughout the year [20]. This requires occupational hygiene expertise, and the information most often includes:

— the job classification of the company;

— the inventory of tasks within a job;

— the task specific exposure profile;

— the operational conditions and risk management measures;

— the duration and location of the exposure within the shift and throughout time, determined by the frequency and period of the tasks;

— experience of the workforce;

— etc.

The SEG concept is developed to simplify the exposure assessment if several workers have the same exposure profile. If the exposure profile is specific to only one worker then the worker's exposure is assessed in the same manner as in the SEG.

If a SEG includes workers from different locations, care shall be taken that the SEG meets the definition 3.1.3 of this European Standard.

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5.2.2 Specifying the measuring procedure

The aim of the measuring procedure and sampling strategy is to obtain valid and representative measurements of the exposure of the workers for comparison with the OELVs taking into account all possible conditions which can reasonably expected to be present throughout time.

Measuring procedures shall comply with the requirements of EN 482. If this is not technically feasible, the reasons shall be given in the report.

NOTE EN 482 refers to further requirements (e.g. on sensitivity, limit of quantification, specificity, capacity of samplers, transportation, stability) in other European Standards such as EN 481, EN 838, EN 1076, EN 13205-1, EN 13890 and EN 13936.

In order to measure the exposure of the worker at the workplace, personal sampling devices shall be used whenever possible, attached to workers' clothing within the breathing zone; workers should be informed about the purpose of the measurement and special recommendations about behaviour during the sampling time. Due to spatial variability of the concentration of chemical agents in workplace atmospheres, static sampling is generally less representative of worker exposure.

The sampling duration is also an important factor that can influence the representativeness of exposure measurements. The sampling duration may exceed the OELV reference period if it is necessary to obtain valid and representative measurements in case of extended workshifts (>8 h).

The sampling duration should be chosen to represent the exposure for the task/activity and representatively describe the exposure for the reference period which is assessed (see Annex D). For this purpose the performance characteristics of the measuring procedure shall be taken into consideration.

The sampling duration should be as close to the OELV reference period as it is necessary to determine the average exposure for the reference period. If the total sampling duration is less than the OELV reference period, then the unsampled time should be assessed carefully and critically for exposure changes that have occurred in the unsampled period and that cannot be ignored. Examples of such situations are presented in Annex D.

For testing compliance with OELV-8 h, three cases shall be considered:

— If exposures occur throughout the workshift and the appraiser considers that the workplace factors, including the tasks, are constant during the workshift, the total sampling duration may be shorter than the reference period with a minimum of 2 h;

— If the appraiser considers that the workplace factors, including the tasks, are not constant during the workshift, the total sampling duration shall be longer than 2 h and as close as possible to the shift duration;

— If the duration of exposure within the workshift is less than 2 h, the sampling duration should cover the whole period of exposure.

When testing compliance with short-term OELV, the sampling duration shall be 15 min.

Measurements shall be performed on sufficient days and during various specific operations in order to gain insight into the pattern of exposure. It is important to consider different episodes during which exposure conditions can vary (night and day cycles, seasonal variations).

The minimum number of measurements shall be in accordance with 5.5.2 and 5.5.3. The appraiser shall also take into consideration the number of workers belonging to the SEG in order to determine the number of measurements that shall be conducted, to ensure that the results are representative of the whole SEG. This is especially important if workers in a SEG belong to different locations.

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The workers chosen for exposure measurement, and the times of measurement, shall be selected so that the measured exposures are representative of the exposure of the workers in the SEG, and this means that it can be necessary to make the measurements at different times. If the number of measurements to be made is greater than the number of workers in the SEG, it will be necessary to measure some workers more than once. As far as possible, this shall be done to make the measurements representative of the exposures in the SEG.

5.3 Performing exposure measurements

Before starting the survey, the appraiser shall make contact with the employer and should ensure that workers are informed about the objectives of exposure measurements. The appraiser shall check that the work activity is consistent with that taken into account in the sampling strategy. The appraiser shall monitor the success of the sampling operation by remaining on site or arranging for it to be done by another person who is sufficiently trained and experienced to monitor the operation effectively. For each sample, the appraiser shall document the following relevant information for the future interpretation of the results of exposure measurements [15]:

— identification of the SEG and of the workers;

— description of the workplace;

— chemical agents relevant for the activity;

— work tasks performed;

— daily working hours and the duration of exposure to chemical agents considered;

— RMM such as engineering control (including their operational status), PPE, work organization, etc.;

— relevant environmental conditions at the workplace (for example temperature, relative humidity, atmospheric pressure, air movement);

— interferences by nearby work activities or others (e.g. smoking);

— presence of other or unusual activities, incidents, etc.;

— sampling information, such as identification of samples and technical sampling equipment used, sampling flow rate, start and end of sampling, etc.

5.4 Validation of results and SEGs

5.4.1 General

Before testing compliance with the OELV it is necessary to consider the validity of each measurement and to use the measurements to evaluate the constitution of the SEG for testing compliance as explained in 5.5.

5.4.2 Validation of measurement results

When the measurement results are available, each measurement shall be evaluated using the information collected during measurement, and by comparison with the other measurements of the SEG.

If the result is unusually high or low, the possibility that this is because of errors of sampling or analysis shall be considered, and if this is the case then the result shall be excluded. However, before excluding a measurement, appraiser notes, and/or contextual information shall be used to identify measurements

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for which accidents with equipment or malfunctions, or deliberate misuse have occurred in relation to the activity, sampling or analysis. If no such irregularity can be reliably identified, the measurement shall not be excluded. If measurements are excluded, it may be necessary to repeat measurements to have sufficient representative measurements to compare with the OELV.

Any removal of results and the reasons shall be explicitly described in the report.

5.4.3 Validation of SEGs

After results have been accepted, they shall be used to validate the constitution of each SEG. Based on expert judgement and information collected during the measurements and the basic characterization, the appraiser shall be able to confirm the constitution of SEGs. When the number of valid measurements is at least 6, the distribution of results shall be examined.

Exposure measurements are usually log normally distributed. To check whether all individuals belong to the same SEG, and whether the distribution of measurements is log-normal, the log probability plot of the data shall be analysed. Guidance is given in Annex E.

An important complication is that two workers doing the same job may not have the same exposure. This phenomenon, known as between-worker variability, means that measurements of one worker's exposure cannot simply be assumed to apply to others doing nominally the same job. Furthermore, exposure varies from shift to shift, giving rise to within-worker variability.

If the measurements show that one or more workers have exceptional exposure, which seems inconsistent with the distribution of results from the rest of the SEG, the reasons shall be investigated and these workers may be treated separately, for example forming a new SEG and taking more measurements as necessary. Guidance is given in Annex E. It may be necessary to repeat exposure measurements to have sufficient representative measurements for each SEG to test compliance.

Any changes to the SEG as a result of these tests shall be recorded in the report (see Clause 6).

5.5 Comparison of results with OELVs

5.5.1 General

5.5.2 and 5.5.3 describe the comparison with an OELV for a single chemical agent. In case of activities involving exposure to several hazardous chemical agents, the EU directive 98/24[28] specifies that the risk shall be assessed on the basis of the risk presented by all such chemical agents in combination. For these situations an exposure index can be calculated from the exposures and the OELVs as described in Annex C. The value of the index shall be compared with 1 instead of comparing the exposure with the OELV. The exposure index is the sum of the exposure of each chemical agent present divided by its OELV:

$$I = \sum_{i=1}^{n} \frac{E_i}{\mathrm{OELV}_i} \qquad (1)$$

where

E_i is the exposure to chemical agent i.

The appraiser shall choose between different approaches mentioned in Annex C.

In the case of extended work shift, the daily exposure (E_d) is used for comparison with the OELV (see Annex G).

This European Standard does not take into account the use and effectiveness of RPE in testing compliance with the OELV.

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5.5.2 Preliminary test

The preliminary test requires three to five valid exposure measurements (see 5.4) on workers belonging to a SEG.

a) If all results are below:

   1) 0,1 OELV for a set of three exposure measurements, or

   2) 0,15 OELV for a set of four exposure measurements, or

   3) 0,2 OELV for a set of five exposure measurements,

   then it is considered that the OELV is not exceeded: **Compliance**.

b) If one of the results is greater than the OELV, it is considered that the OELV is exceeded: **Non-compliance**.

c) If all the results are below the OELV and one result is above 0,1 OELV (set of three results) or 0,15 OELV (set of four results) or 0,2 OELV (set of five results) it is not possible to conclude on compliance with the OELV. **No-decision.** In this situation additional exposure measurements shall be carried out (requiring at least at total of six measurements) in order to apply the test based on the calculation of the confidence interval of the probability of exceeding the OELV, as specified in 5.5.3.

According to EN 482, the measuring range for short-term exposure measurements may be from 0,5 to 2 times the OELV. In this case, the preliminary test cannot be used for short-term measurements (reference period equal to 15 min).

5.5.3 Statistical test

The appraiser shall select a statistical test to check whether the exposures of the SEG comply with the OELV. The test shall measure, with at least 70 % confidence, whether less than 5 % of exposures in the SEG exceed the OELV.

A suitable test is given in Annex F. Other tests may be used provided that they have been shown to meet the above confidence specification.

The appraiser shall make exposure measurements of the SEG according to the procedures in 5.2.2 and 5.3, in a way which meets the requirements of the test chosen. The results shall be analysed statistically according to that test.

If one or more of the exposure measurements are below the limit of quantification (LOQ), and the statistical test selected involves use of geometric (GSD) or standard (SD) deviations and/or geometric (GM) or arithmetic (AM) means, then the values below the LOQ shall be treated in a way which produces a reliable result. Methods are described in Annex H for a case where a minority of results are < LOQ.

Applying such tests to the exposure measurements assumes that the measurements are log-normally distributed (see Annex E). This is usually true at least approximately, and it is unusual to have enough measurements to show statistically that it is not true.

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6 Report

Reports shall be written of the occupational exposure assessment and of any periodic measurement.

The report shall contain the following information:

— name of the appraiser and institutions undertaking the assessment and the measurements;

— purpose of the assessment;

— name of the chemical agents considered;

— name and address of premises;

— description of the workplace factors and working conditions;

— observations made during sampling;

— results and conclusions of the basic characterization;

— measuring procedure and equipment used and agreement with requirements of EN 482 (see Note in 5.2.2);

— time schedule (date, beginning and end of sampling);

— exposure concentrations;

— details of quality assurance (e.g. expanded uncertainty according to EN 482);

— clear identification of results;

— result of the comparison with the limit value.

If RPE is used, the results of measurements do not correspond to the concentration inhaled by the workers and this shall be reported.

In case of non-compliance, the appraiser may suggest the reasons and the risk management measures to remedy the situation.

7 Periodic reassessment

[AC〉 Occupational 〈AC] exposure assessment shall be updated periodically to keep it current and ensure that worker's exposure remains in compliance with the OELV. In case of significant changes in workplace factors the basic characterization shall be updated immediately.

Even if the risk assessment for inhalation exposure shows that RMM are adequate (RMM evaluation) and/or exposure is below the OELV (compliance), changes affecting exposure, such as gradual deterioration in ventilation equipment, or subtle changes in ways of working can occur without the workers or appraisers being aware. Thus, despite the absence of significant modification of workplace factors, the appraiser shall justify why also in future compliance can be expected.

Reassessment of exposure can be done with exposure measurements or other methods (see Annex A).

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In general, an annual interval is recommended for reassessment, whatever methods used.

When reassessment is conducted with exposure measurements, periodic intervals for measurements are proposed in Annex I.

For periodic reassessments with measurements, the appraiser shall decide upon the number of periodic measurements per SEG. The results shall be validated according to 5.4. If the results pass the validation, these results can be pooled with the earlier ones in order to test compliance and to determine the period before the next reassessment.

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Annex A
(informative)

Assessment of exposure

A.1 General

This annex gives guidance on when measurements are advisable or if other approaches of assessment may be used.

In practice there are different ways of assessing workers´ exposure to chemical agents. This annex is intended as a support to the identification and assessment of inhalation exposure on workplaces.

In assessment, it may be necessary to consider together all activities in a working area, which is the spatially or organisationally delimited part of a site in which activities involving chemical agents are carried out by one or more workers and can be brought together in a risk assessment. It may encompass one or more workplaces or working processes. The main factors to be used to choose the appropriate method for assessing exposure are extensive, including the type and duration of exposure by inhalation at workplaces.

In addition to exposure measurements, the methods used to assess exposure generally include:

— reasonable worst case measurements (under control);

— measurement of technical parameters, e.g. air velocity, air change rate);

— calculation of exposure (using appropriate models or algorithms);

— comparison with other workplaces, in the same enterprise or in other enterprises;

— control banding approaches, and

— good practice guidance for defined branches or tasks.

Reasonable worst case measurements are performed to estimate the maximum exposures likely to occur with the controls in place.

When applying one of these methods and there is still uncertainty concerning the identification result and its evaluation, additional workplace measurements are recommended. In A.2 to A.9 different workplace situations are described together with information about possible methods to assess exposure and to derive suitable protective measures. Table A.1 gives an overview of workplace situations addressed in this Annex and the recommended way of exposure assessment.

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Table A.1 — Overview of approaches to exposure assessment in different workplace situations

Workplace situation	Exposure measure-ments (according to Clause 5)	Reasonable worst case measurements (under control)	Measurement of technical parameters	Calculation of exposure (using validated models or algorithms)	Comparison with other workplaces	Control Banding approaches	Good practice guidance for defined branches or tasks
A.2 constant conditions	x	x	x	x	x	x	x
A.3 shortened exposure with constant conditions	x	x	x	x	x	x	x
A.4 occasional exposure	x	x	x	x	x	x	x
A.5 stationary with irregular exposure	x	x	x	x	x	x	x
A.6 mobile with irregular exposure	x	x	-	x	x	x	x
A.7 unpredictable, constantly changing exposure	x	-	-	-	x	-	x
A.8 outdoor	x	x	x	x	x	-	x
A.9 underground	x	x	x	x	x	-	x

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A.2 Workplaces with constant conditions

a) What is typical for workplaces with constant conditions is the fact that day after day, month after month the same activities are carried out under recurring conditions. Everything that influences the workers´ exposure to chemical agents changes little over a long period. Work sequences, technologies used, process parameters (e.g. pressure, temperature), type and quantity of the chemical agents used, degree of utilization of the installation and ambient conditions (ventilation, climate) are repeated in every shift and remain relatively constant for the long term.

b) Examples of workplaces with constant working conditions include industrial serial production, print shops, dry cleaning, commercial sterilization, quality controls in routine laboratories or large-scale chemical plants.

c) It is characteristic for such workplaces that:

   1) the exposure is typically recorded in the form of the shift average values;

   2) the workplace factors which affect exposure are repeated regularly, and

   3) the exposure profile changes little in the long term.

d) At these workplaces basically all assessment methods may be used. Where measurement results from similar workplaces are available, or results from exposure modelling, these may be used as guides.

A.3 Shortened exposure at workplaces with constant working conditions

a) Frequently there are workplaces where certain activities involving chemical agents are performed for only a short time every day. In the remaining time other activities (not involving exposure to chemical agents) are carried out. This type of occupation is repeated daily and also remains unchanged over a period of months.

b) Examples of such workplaces are cold sterilization in hospitals, stationary welding work or sampling operations for a limited time.

c) Everything that influences the exposure of the workers to chemical agents in the period which is shorter than the length of the shift changes little overall. In every shift it is possible to distinguish clearly between periods with exposure to chemical agents and periods without exposure to chemical agents.

d) The working sequences involving exposure to chemical agents are assessed as described in A.2 if the conditions described are met.

e) Even when the occupational exposure limit is complied with it can be appropriate to provide for additional protective measures precisely during the short periods.

A.4 Workplaces involving occasional exposure

a) If the activities involving exposure to chemical agents by inhalation at stationary workplaces only arise occasionally, regular workplace measurements are difficult to plan. At such workplaces certain working procedures or activities with a comparable exposure to chemical agents are repeated irregularly, e.g. every week, every month or even more seldom (occasional batches).

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b) It is typical of such workplaces, for example, that they can involve changing batches in the chemical industry or maintenance work on permanently installation equipment.

c) The inhalation exposure is assessed separately for each individual case where there are different activities and working procedures within the framework of the identification and assessment. Unless there is other information, base the assessment on cases where the most unfavourable exposure conditions apply.

A.5 Stationary workplaces with irregular exposure

a) Assessment of the exposure to chemical agents in working areas where activities and the use of chemical agents are changing currently is particularly difficult.

b) This can be the same, for example, in craft companies, for non-specialized maintenance, repair, production and assembly work, in automotive or forklift workshops, for services at filling stations (oil change etc.) or in research laboratories.

c) At such workplaces certain working procedures or activities with comparable exposure to chemical agents are repeated irregularly, e.g. every day, every week, every month or even less often. Other or comparable activities involving varying exposure to chemical agents may occur during the shift which can influence the measurements.

d) The identification of the inhalation exposure for the various activities is laborious, especially with respect to the measurements. But since the activities and the materials used within a sector mostly differ only insignificantly from company to company, cross-company surveys and assessments from e.g. other companies or surveillance authorities, practical instructions relating to good practice or process- and substance-specific criteria, are the method of choice.

A.6 Workers who move from a workplace to another with irregular exposure

a) The procedure described in this European Standard can encounter its limits, however, where work is performed at locations which change a number of times a day.

b) This normally applies to activities performed on construction sites (e.g. painters, floor-layers, bituminous pavers) or non-stationary assembly and maintenance work (e.g. on photocopiers).

c) In such activities the individual working steps normally follow the same sequence, while the number of operations, the procedures and materials applied and the scope of work may vary. Additionally, the ambient conditions (ventilation, room size and work in the open) vary considerably.

d) Here the working procedures and the use of materials should be designed from the outset in such a way that risks due to inhalation exposures do not even arise.

e) All these measures assume that the exposure situation for the respective activity has been identified and assessed beforehand on a sector basis. With the application of “safe working procedures” care should be taken to ensure that no additional risks can arise due to the external environment (background exposure etc.).

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A.7 Workplaces with unpredictable, constantly changing exposure

a) There are workplaces where everything can change, which will influence the exposure to chemical agents. For example, with many jobs in contaminated areas (e.g. investigation and remediation of existing contamination, construction work at landfills or any sites contaminated accordingly by industrial or commercial use, and with the demolition of industrial plants) or at special waste collection facilities it is for the most part not reliably known what chemical agents will occur and in what form.

b) Prior investigations can only give indications of existing chemical agents and their distribution. The reliability of these indications depends among other things on the correct selection of the investigation parameters for the specific location, the type and density of sampling etc., i.e. the representative nature of the investigations to be assumed for the object of the investigation (subsoil, groundwater, brickwork etc.). At locations such as landfills for household waste the stock of chemical agents actually present can also hardly be described reliably by means of extensive samplings and corresponding analyses. Furthermore there is no fixed workplace (see A.8). Even if the nature of the work remains constant, highly fluctuating exposures can arise due to the (unforeseen) occurrence of different chemical agents not recorded in the investigations.

For this reason the chemical agents exposure of workers can normally only be ascertained in real terms in retrospect (the use of biomonitoring can provide helpful information). Protective measures based on a risk assessment should be applied for such work, measures which take account of the reasonable worst case.

A.8 Outdoor workplaces

In the case of work conducted outdoor the identification and assessment of exposure by measurements according to the specifications of this European Standard should be done more carefully because of the variability of exposure. This includes, for example, activities in stone quarries, agriculture and forestry, horticulture, road-building, at coking plants and open-air facilities in the chemical industry, during the loading and emptying of tankers, during roofing work or when coating facades.

Assessment of work out of doors has the additional difficulty of varying weather conditions, the possibility of very different conditions of work should be considered in the assessment, and if SEGs are formed the effect of weather on work practice should be carefully considered.

An indication of the potential exposure can be obtained by measurements conducted in the proximity of the emission source.

A.9 Underground workplaces

a) At underground workplaces, e.g. load work in potassium or salt mining, work at the face and in the roadways of coal mines, drilling work, road heading, maintenance and repair work and work on mine supports, identical or similar categories of work are carried out under changing ambient conditions which are nevertheless always relatively well known (ventilation). Chemical agents which are released will under certain circumstances also influence mine areas which are downstream in ventilation terms.

b) Conditions of work are also changed by the very act of mining and today's measurements not be representative for tomorrow's conditions, as the geometry of the respective site will most probably have been altered substantially.

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c) Measurements conducted within the framework of the identification and assessment of inhalation exposure and control measurements are therefore sometimes difficult. However, it is frequently possible to conduct reliable calculations of the concentrations on the basis of substance-specific and process-specific data (e.g. emission rates/release rates, ventilation data) and the parameters of the technical procedure in use.

d) The sustainable guarantee of the effectiveness of the protective measures should be ensured by means of technical (e.g. ventilation) or organizational (e.g. quantity limitation) measures.

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Annex B
(informative)

Occupational exposure limit values for compliance testing

This annex gives an overview of the different types of occupational exposure limit values (OELVs) which can be used for testing their compliance in relation with this European Standard.

In the scope of this European Standard an OELV is specified pursuant the definition in EN 1540 as limit of the time-weighted average of the concentration of a chemical agent in the air within the breathing zone of a worker in relation to a specified reference period, usually 8 h for long-term measurements and 15 min for short-term measurements.

If there is exposure to more than one chemical agent an exposure index should be calculated from the exposures and the OELVs as described in Annex C. The value of the index should be compared with 1 instead of comparing the exposure with the OELV, as described in 5.5.1 of this European Standard.

Depending on the goal of the assessment, an appraiser can determine which of the following – whether short-term or long-term - OELVs should be used to test its compliance:

- legal occupational exposure limits, e.g.:
  - European binding occupational exposure limit values;
  - National binding occupational exposure limit values.
- recommendation of scientific committees, e.g.:
  - indicative OELVs;
  - maximum allowable concentrations;
  - recommended exposure limits;
  - threshold limit values.
- exposure limits value provided by the supplier:
  - derived no-effect levels or derived minimum effect level for the inhalation route.
- technical derived limit values or risk related concentration;
- risk-based limit value:
  - acceptable or tolerable risk related concentration;
  - technical risk related concentration.
- action values;
- in-house limit values.

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Note that legally binding OELVs are often deduced from one of the other types of OELVs, e.g. recommendations of scientific committees or technical derived limit values. Recommendations of scientific committees are often health-based.

Derived no-effects levels (DNELs) or derived minimum effect level (DMELs) may be provided by manufacturer/distributor in the safety data sheet.

Action levels and in-house limit values are often derived to meet certain demands and can consist of any type of the before mentioned types of OELVs as well as compute by an expert of the company itself, e.g. as kick-off value.

Examples of sources for the different OELVs are:

— SCOEL website with BOELV and IOELV:

  http://ec.europa.eu/social/main.jsp?catId=148&langId=en&internal_pagesId=684&moreDocuments=yes&tableName=INTERNAL_PAGES

— IFA website - GESTIS database on hazardous substances:

  http://www.dguv.de/ifa/GESTIS/GESTIS-Stoffdatenbank/index-2.jsp

— IFA website - GESTIS international limit values:

  http://limitvalue.ifa.dguv.de/

  http://www.dguv.de/ifa/gestis/gestis-dnel-datenbank/index-2.jsp

— List of MAK and BAT Values of the German Permanent Senate Commission for the Investigation of Health Hazards of Chemical Compounds in the Work Area:

  http://onlinelibrary.wiley.com/book/10.1002/3527600418/homepage/access_to_the_list_of_mak_and_bat_values.htm - DOI: 10.1002/9783527805983

— Occupational Exposure Limits (OELs) for substances in the Netherlands:

  https://www.ser.nl/en/oel_database.aspx

— Comprehensive database for Occupational Hygiene Exposure Limits:

  https://www.dohsbase.nl/en/

— List of occupational exposure limits of the International Social Security Association (ISSA)

AC — List of French OELs for chemical agents

  http://www.inrs.fr/media.html?refINRS=ED%20984 AC

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Annex C
(informative)

Simultaneous ▌AC⟩ occupational ⟨AC▐ exposure to several chemical agents

C.1 General

Exposure to several chemicals agents during the work-shift occurs at many work places. Some examples of this are in painting operations, welding of metallic alloys, sanding, handling insulations, cementing, underground work etc.

Although knowledge of the health effects of such mixtures in many cases is limited, multiple exposures should be taken into account to characterize the situation, in particular for comparison with occupational exposure limit values (OELVs). For this, several approaches exist, ranging from simple exposure index methods to advanced toxicological models.

Care should be taken to ensure that exposures to several chemical agents both individually and in combination are properly addressed. For assessment of several exposures to different chemical agents, a tiered approach as described below can be applied:

— Tier 1: calculate an exposure index (I_E) as in 5.5.1, i.e. the sum of the exposures to all the chemical agents divided by their respective OELVs;

— Tier 2: calculate an exposure index (I_{AE}) i.e. the sum of the exposures to all the chemical agents with similar effects or acting on the same target organ divided by their respective OELVs;

— other toxico-kinetic based methods (not described in this European Standard).

For comparison with the OELV, as in 5.5.1, the Tier 1 exposure index is calculated for all measured compounds at the work place regardless of its physical state or possible health effects. The Tier 2 exposure index assumes that only the health effects of the chemical agents (or groups of chemical agents) with similar effects of the same target organ are additive.

These two methods are in most cases conservative, however when multiplicative or synergetic effects exist, both Tier 1 and Tier 2 might underestimate the risk related to the exposure. In these cases, other methods should be used.

Due to the complexity of the matter, the use of other and less conservative approaches than Tier 1 and Tier 2 should be justified. After the exposure index (I_E or I_{AE}) has been calculated they should be further treated as described in 5.5.1.

C.2 Tier 1: Exposure index (I_E)

If the appraiser has no information about the combined effects, this approach should be used.

According to Formula (C.1) the simultaneous exposure index, I_E, is the sum of the exposures to each of the component in the workplace atmosphere divided by their respective $OELV_i$.

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The OELV to be applied to the simultaneous exposure index is 1.

$$I_E = \sum_{i=1}^{n} \frac{E_i}{\mathrm{OELV}_i} \tag{C.1}$$

where

n is the number of chemical agents the worker is exposed during the workshift.

Example C.1

Following this approach, an exposure index (I_E) can be calculated for an exposure occurring from rubbing down with abrasive paper, and cleaning with solvents, where the worker was exposed to inhalable dust and mixture of solvents (2-methoxyethyl acetate, cyclohexanone and toluene).

The OELV, the time weighted average (TWA-8 h) and the OELV fraction for the chemical agents are presented in Table C.1.

Table C.1 — Several exposures to chemicals agents (Agent (i), OELV, TWA and the OELV fraction for the chemical agents)

Chemical agent	**Agent (i)**	**OELV-8 h** mg/m³	**Results (TWA)** mg/m³	$\frac{E_i}{\mathrm{OELV}_i}$
Dust (inhalable)	1	10	1,2	0,12
2-methoxyethyl acetate	2	4,9	1,8	0,37
Cyclohexanone	3	40,8	11	0,27
Toluene	4	192	38,4	0,20

$$I_E = \frac{E_1}{\mathrm{OELV}_1} + \frac{E_2}{\mathrm{OELV}_2} + \frac{E_3}{\mathrm{OELV}_3} + \frac{E_4}{\mathrm{OELV}_4}$$

$$I_E = \frac{1,2}{10} + \frac{1,8}{4,9} + \frac{11}{40,8} + \frac{38,4}{192}$$

$$I_E = 0,96$$

C.3 Tier 2: Additive effect exposure index (I_{AE})

According to Formula (C.2) [25] this additive effect exposure index, I_{AE_j}, is the sum of the exposure fractions of the component with similar effects or acting on the same target organ (C.2). The OELV to be applied to the additive exposure is 1.

$$I_{AE_j} = \sum_{i=1}^{m} \frac{E_i}{\mathrm{OELV}_i} \tag{C.2}$$

where

m is the number of chemical agents with similar health effects or target organ the worker is exposed to.

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Example C.2

This is the same example as previous, where the chemical agents have been grouped by their health effects (*j*), dust lung (j_1), irritating effects (j_2) and reproductive effects (j_3), respectively, due to different critical effects and target organs (see Table C.2).

Table C.2 — Several exposures to chemical agents (Agent (i), Effect group (j), OELV, TWA and the OELV fraction for the chemical agents)

Chemical agent	Agent (i)	Effect group (*j*)	OELV-8 h mg/m³	Results (TWA) mg/m³	$\frac{E_i}{\mathrm{OELV}_i}$
Dust (respirable)	1	A	10	1,2	0,12
Cyclohexanone	2	B	40,8	11	0,27
2-methoxyethyl acetate	3	C	4,9	1,8	0,37
Toluene	4	C	192	38,4	0,20

$$I_{\mathrm{AE}_A} = \frac{E_1}{\mathrm{OELV}_1} = \frac{1,2}{10} = 0,12$$

$$I_{\mathrm{AE}_B} = \frac{E_2}{\mathrm{OELV}_2} = \frac{11}{40,8} = 0,27$$

$$I_{\mathrm{AE}_B} = \frac{E_3}{\mathrm{OELV}_3} + \frac{E_4}{\mathrm{OELV}_4} = \frac{1,8}{4,9} + \frac{38,4}{192} = 0,37 + 0,20 = 0,57$$

C.4 Other approaches

Although knowledge of the health effects of mixtures is limited, some knowledge exists for assessing combined exposure from different chemical agents. When such exposure situations occur, the assessor is advised to consult literature to check if any more specific assessment approach exist. These approaches might include variations of exposure index methods[25], approach focussing on assessing the critical or lead component in a mixture – the Critical Component Approach (CCA), method described in the Dangerous Preparations Directive (DPD+) [26], or other advanced toxicological models, including mode of action and kinetic and dynamic aspects of the exposure-effect relationship.

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Annex D
(informative)

Exposure profile and sampling duration

D.1 General

The total sampling duration is specified by the appraiser taking into account the workplace factors including tasks. The sampling duration should be representative for the reference period of the limit value controlled and should be established considering the concentration variability, the LOQ and other performance characteristics of the analytical method.

NOTE Figure D.1 to Figure D.8 give exposure concentration *C* versus time *t*. For example, these traces can be obtained with continuous-reading instruments during the basic characterization (see 5.1.1).

D.2 Measurement for testing compliance with 8 h- OELV

a) Workplace factors are constant during the whole work shift (*WS*)

Total sampling duration (TSD) should be a minimum of 2 h.

The exposure of the sampling period can be considered representative of the exposure of the work shift duration (WS).

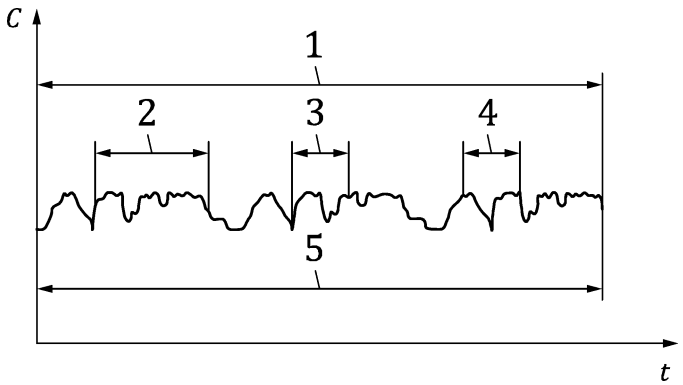
There are three alternatives in the following order of priority:

— Measure for the full shift (TSD_{max});

— Measure for one period of exposure (at least 2 h) (TSD_{min});

— Measure for more than one period of exposure (i.e. two periods of 1 h; TSD_{min}).

See Figure D.1.



Key

C	exposure concentration	2	TSD_{min} = 2 h
t	time	3 and 4	TSD_{min} = 1 h+1 h
1	TSD_{max} = WS	5	WS

Figure D.1

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b) Workplace factors are not constant during the work shift

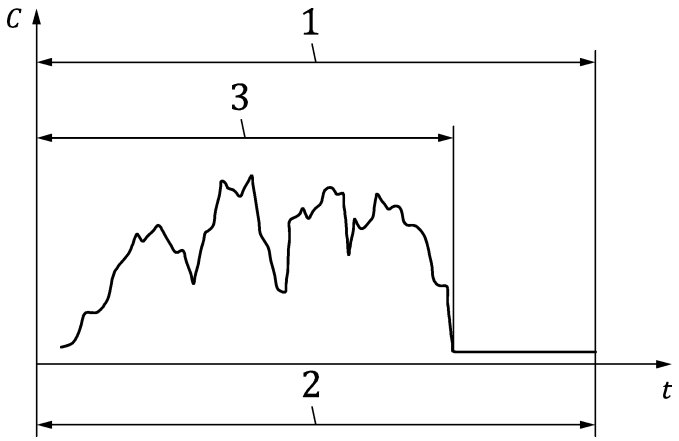
1) Exposure occurs during a part of shift

There are three alternatives:

— Measure for the full shift and use the average exposure for the whole shift (TSD_{max});

— Measure for the total period of exposure (TSD_{PE}), and assume that the exposure for the rest of the shift is zero after careful examination to calculate the 8 h-TWA (For example, this case can be apply when the worker leaves the workplace);

See Figure D.2



Key

C	exposure concentration	1	TSD_{max} = WS
t	time	2	WS
		3	TSD_{PE}

Figure D.2

— Measure for the period of highest exposure (TSD_{HE}) if based on the basic characterization, this period of the highest exposure can be reliably determined. This exposure measurement is assumed to apply for the total period of exposure.

See Figure D.3

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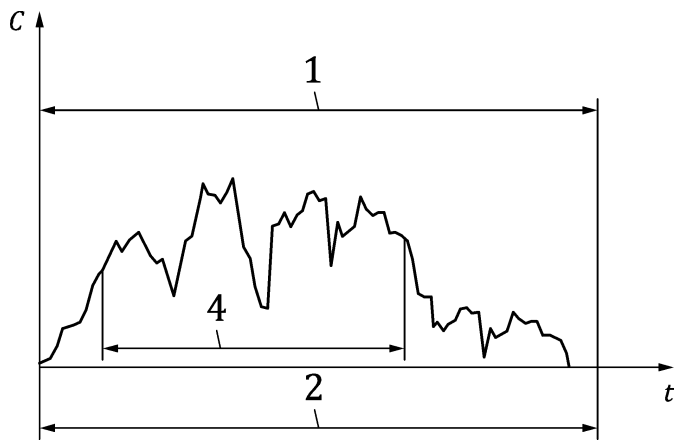

Key

C	exposure concentration	1	TSD_{max} = WS
t	time	2	WS
		4	TSD_{HE}

Figure D.3

2) Multiple exposure scenarios during the whole shift for one chemical agent.

Tasks are randomly distributed during the workshift, the sampling duration should correspond to the shift duration.

There are two alternatives:

— Measure for the full shift (TSD_{max});

— Measure each task and calculate the average exposure over the whole shift (TSD_i).

NOTE Consecutive samples collected during the whole workshift can give valuable information about levels of exposure corresponding to different tasks.

See Figure D.4.

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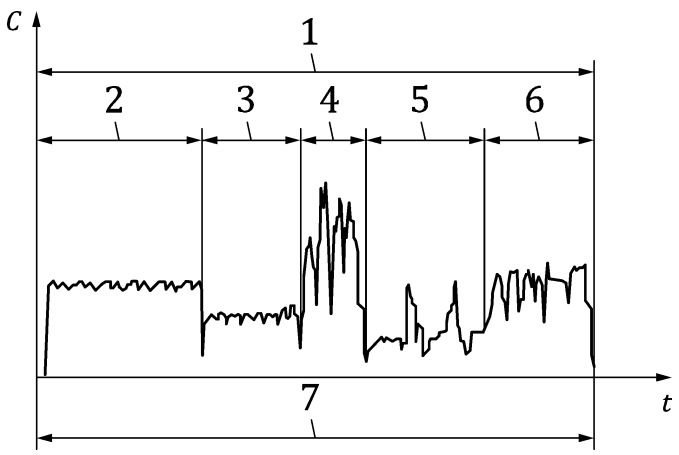

Key

C	exposure concentration	4	TSD_3
t	time	5	TSD_4
1	TSD = WS	6	TSD_5
2	TSD_1	7	WS
3	TSD_2		

Figure D.4

3) Single exposure scenario repeated n times during the whole shift

There are three alternatives:

— Measure for the full shift and use the average exposure for the whole shift (TSD_{max});

— Measure for at least one cycle of the exposure profile or, if a cycle lasts less than 2 h, for at least 2 h and for a complete number of cycles (TSD_{min});

— Measure for the period of highest exposure (TSD_{HE}) if based on the basic characterization, this period of the highest exposure can be reliably determined. This exposure is assumed to apply for the period of exposure.

NOTE In this situation special attention is paid to peaks of exposure (referred to 5.1.1).

See Figures D.5 and D.6.

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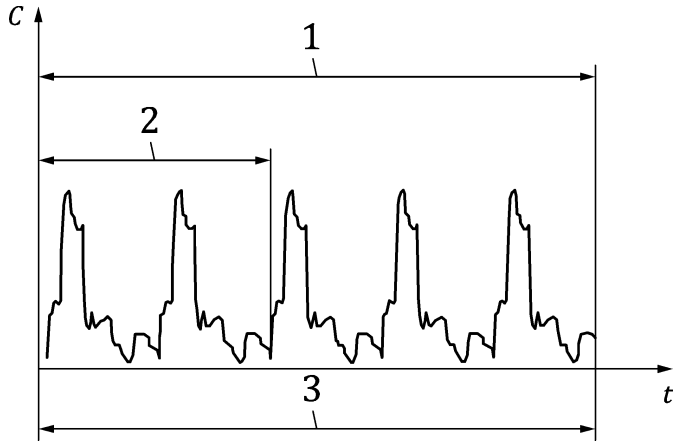


Key

C exposure concentration

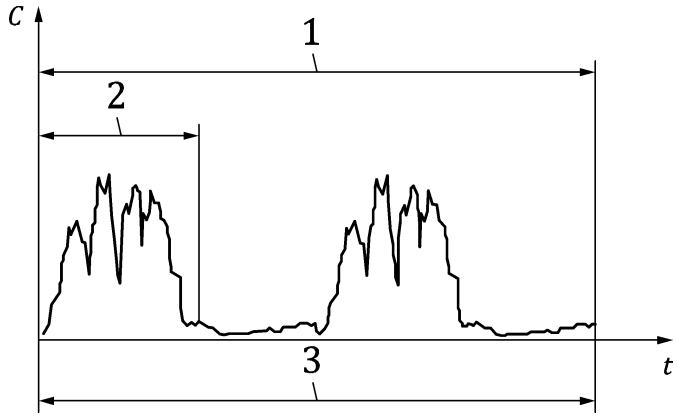
t time

1 TSD_{max} = WS

2 TSD_{min} = whole cycle ≥ 2 h

3 WS

Figure D.5



Key

C exposure concentration

t time

1 TSD = WS

2 TSD_{HE} = 2 h

3 WS

Figure D.6

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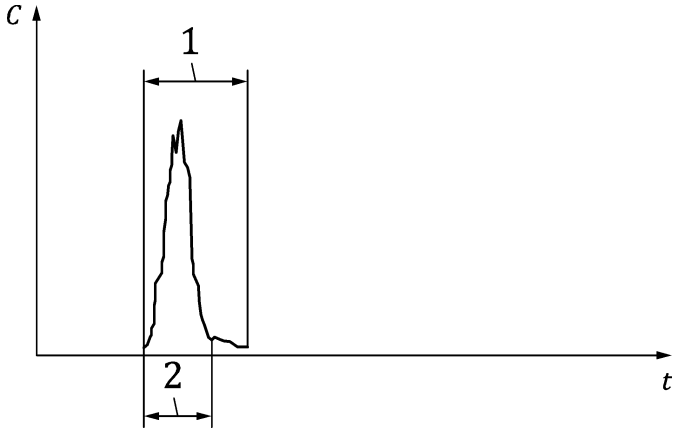
D.3 Measurement for testing compliance with short-term limit value

a) Tasks with a duration shorter than the reference period

The total sampling duration should cover the corresponding reference period (RP) and include the whole task in order to take into account the residual concentration after the task. See Figure D.7.



Key

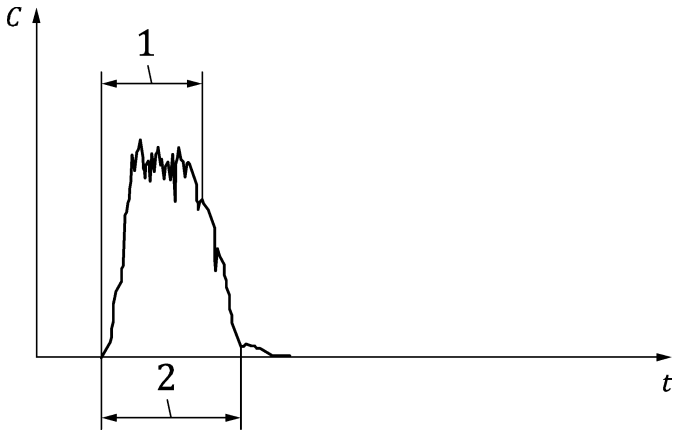
C	exposure concentration	1	TSD = RP
t	time	2	task

Figure D.7

b) Tasks with a duration longer than the reference period

The sampling duration (TSD) is equal to the reference period (RP) and covers the period of highest exposure after careful examination. See Figure D.8.

NOTE If the sampling period is longer than the reference period of the short-term OELV, then the same procedure recommended for OELV 8 h and described in Annex G can be applied.



Key

C	exposure concentration	1	TSD = RP
t	time	2	task

Figure D.8

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Annex E
(informative)

Check of exposure measurements distribution, and identification of exceptional exposure within the SEG

E.1 General

Annex E gives graphical and statistical methods for the validation of the results and the SEGs, as required in 5.4, for the identification of individuals or groups whose exposure is not properly represented by the SEG, and for the decision to use the normal or the lognormal distribution in the compliances test (see Annex F). The graphical method supports the identification of a LOQ (see Annex H) and is applied using at least six exposure measurements in the SEG.

The measurement results may be examined to see if a normal or lognormal distribution fits the results better. In many circumstances, exposure variability is influenced by the multiplicative interactions of workplace factors (see 5.1.3, Annex A and Annex D). Measurements would be expected to approximate to a lognormal distribution, and this has often been observed[19],[20]. However, a normal distribution may appear with high control of workplace factors in constant conditions (see A.2) or if the influence of workplace factors is outshined by the random variation in the measuring procedure (see 5.2.2)[24].

The shape of the measurement results may be examined by a statistical test, such as the Shapiro and Wilk test[3], to see whether their distribution is closer to lognormal than normal distributions. For example, the results may first be tested as to whether their distribution fits a normal distribution. This will give a probability p_1 of the hypothesis that the normal distribution fits the results. The test may then be repeated using the logarithms of the results, giving a probability p_2. If $p_1 > p_2$ it may be concluded that a normal distribution fits the results better than a lognormal distribution. The result of comparing p_1 and p_2 determines the distribution to be used for the compliance test (see Annex F).

The appraiser should take into account that the exposure distribution may deviate in the tails from the (log)normal model (see E.2.3) and that the normal distribution does not exclude negative concentrations. Truncation at higher concentrations (vapour saturation, explosion limit or maximum dustiness) can occur, like censoring in the lower tail due to the increasing number of results below the LOQ. Therefore the check on the distribution and on pooling results are important before LOQ adjustment (see Annex H) and applying the models in the compliance test (see 5.5.3 and Annex F).

E.2 describes a graphical method for a lognormal distribution. To decide if a normal distribution describes the results better, the procedure should be repeated using normal probability paper.

E.3 refers to software that automates the graphical method of E.2, test goodness-of fit (see E.2.2) and test the consistency of the SEG (see E.2.4.3).

E.2 Graphical method

E.2.1 Principle

The graphical method involves plotting the measured values of exposure x in ascending order on log-probability paper. Normal probability paper can also be used if log-probability paper is unavailable. In this case, $\log(x)$ is plotted instead of x.

The plot is then examined as described below.

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E.2.2 Plotting

The n measured values of exposure x_k are arranged in ascending order (x_1, x_2, x_3, ... x_n). These values are plotted on the horizontal axis against the corresponding probabilities P_k on the vertical axis, where

$$P_k = (k - 3/8)/(n + 1/4) \quad \text{(E.1)}$$

Formula (E.1) is specified by ISO 5479. It calculates P_k as a decimal with a maximum value of 1. Commercial probability paper often presents P_k as a percentage, in which case the values of P_k calculated from Formula (E.1) are multiplied by 100 before plotting.

E.2.3 Example

Nine exposure measurements were made in a SEG, with the following results, arranged in ascending order: 0,32; 0,60; 0,62; 0,90; 0,93; 1,1; 1,2; 1,35; 2,4 (all in mg/m^3).

Table E.1 shows these x_k values, their corresponding ordinal values k, and the corresponding cumulative probability values P_k for plotting, calculated from Formula (E.1). Figure E.1 shows these values plotted in a log-probability diagram.

Table E.1 — Example of nine exposure measurements with the associated probability values for plotting in a log-probability diagram

Exposure x_k mg/m^3	k	P_k	P_k **as percentage**
0,32	1	0,068	6,8
0,60	2	0,176	17,6
0,62	3	0,284	28,4
0,90	4	0,392	39,2
0,93	5	0,500	50,0
1,1	6	0,608	60,8
1,2	7	0,716	71,6
1,35	8	0,824	82,4
2,4	9	0,932	93,2

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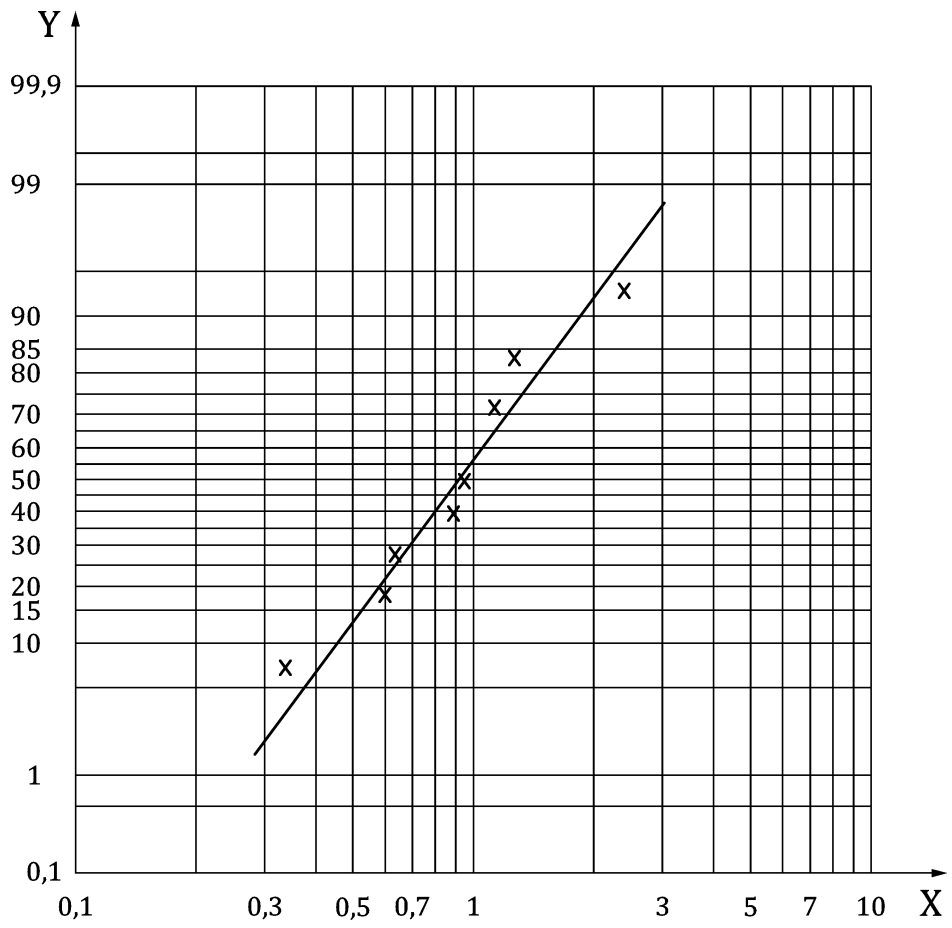

Key

X exposure x_k (mg/m³)

Y probability P_k × 100

Figure E.1 — Nine exposure measurements and their probability values plotted in a log-probability diagram

The good fit to a straight line shows that these results are distributed lognormally.

In many cases, such as Figure E.1, it is clear that the log-probability plot is close to linear, and it can be concluded that for practical purposes it is acceptable to assume that the distribution of exposure measurements is lognormal. In the words of ISO 5479, "If a plot on this paper gives a set of points that appears to be scattered around a straight line, this provides crude support for the assumption that the sample can reasonably be regarded as having come from" the distribution tested. Also, ISO 5479 indicates that the variance of the values at the ends of the distribution are greater than those in the middle, so points at the ends can be further from the line than in the centre. Interpretations of deviations from the straight line can be found in public available literature[24].

E.2.4 Examples of SEGs requiring further consideration

E.2.4.1 General

E.2.4.2 to E.2.4.4 gives examples of plots of data sets which suggest that the assumption of the lognormality of the data and conformity of the SEG to definition 3.1.3 are in doubt. Definite conclusions

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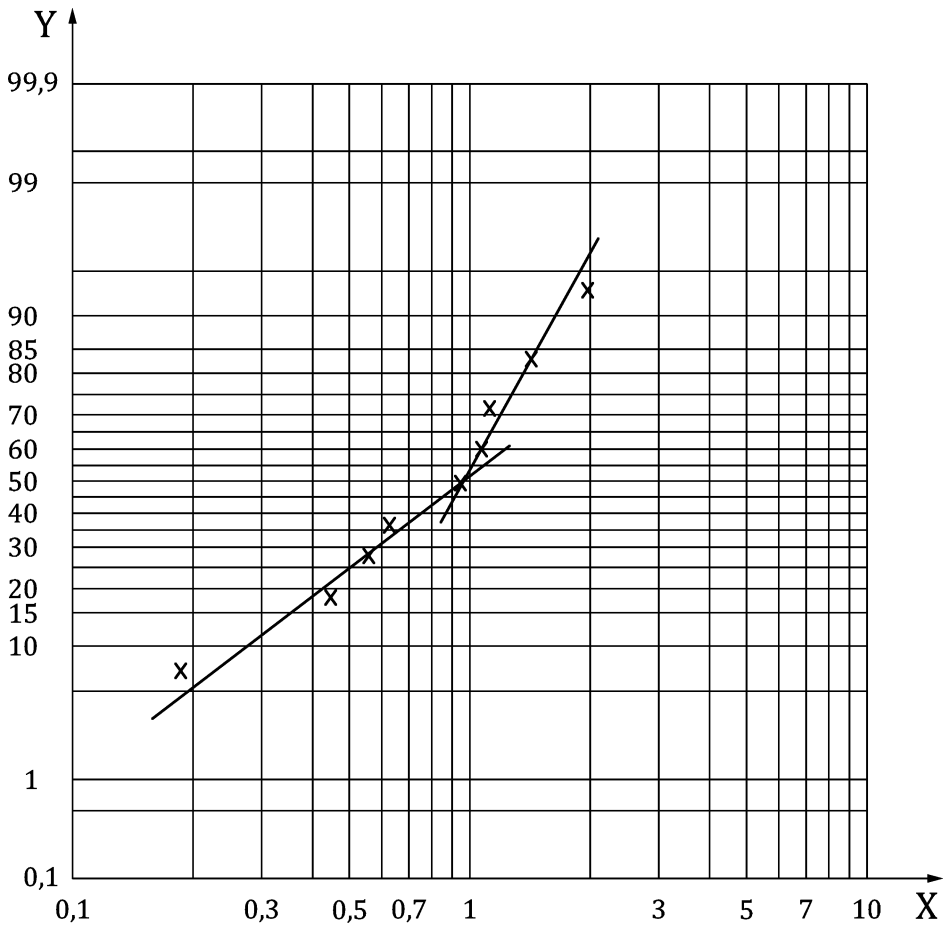
cannot be drawn from such plots with small numbers of data points. They can only indicate that the Basic Characterization (see 5.1) and Constitution of SEGs (see 5.2.1) should be reconsidered.

E.2.4.2 SEG consisting of two groups

Figure E.2 illustrates a log-probability plot which appears to be two intersecting straight lines rather than a single one. These measurements were divided into two different groups (with the central point in each, as this result could be in either group), and the two sets plotted separately on the same chart in Figure E.3.

Figures E.2 and E.3 suggest that the SEG includes two different groups with different exposure patterns, and the SEG should be divided in two, and more exposure measurements made so that the number of measurements in each SEG reaches the minimum required by the statistical test chosen in 5.5.3. If these extra measurements are added to the original plot, modifying Figure E.2, they may clarify whether one or two distributions are present, or if there is some other explanation.



Key

X exposure x_k (mg/m³)

Y probability P_k × 100

Figure E.2 — Example of exposure measurements which appear to come from two different populations within the SEG, each with a lognormal distribution of exposure

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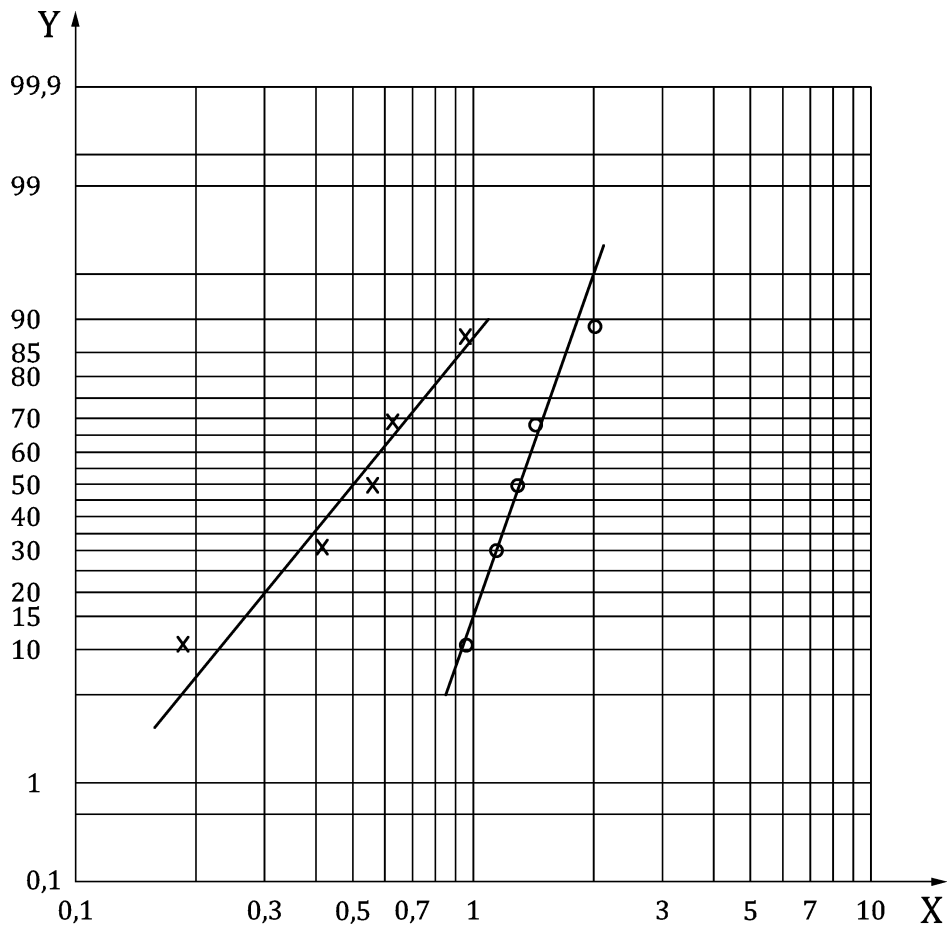

Key

X exposure x_k (mg/m³)

Y probability P_k × 100

Figure E.3 — Exposure measurements replotted as two different sets (with the central point included in both), confirming that these appear to be two different lognormally-distributed populations

E.2.4.3 Some individuals with exceptional exposure

Figure E.4 illustrates a plot in which two of the exposures are much higher than expected from the exposure distribution of the rest of the SEG. It is not possible to say from these results whether this is a chance variation, so that on a different day the exposure of these individuals would fit in with the rest, or whether there is something about their exposure factors or work practices that results in consistently high exposure.

The information obtained by the appraiser should be examined to see if it reveals any reason for the exceptional results (see 5.1, 5.3 and 5.4.2). It can be necessary to observe these workers further, perhaps with the aid of direct-reading instruments, and further measurements of their exposures can be necessary to see if they really are exceptional. If it appears that their exposures are exceptional, these workers should be separated as a different SEG (which may consist of one person) and more measurements made for their SEG, and for the one they were in, to make up the right number of measurements for the statistical test.

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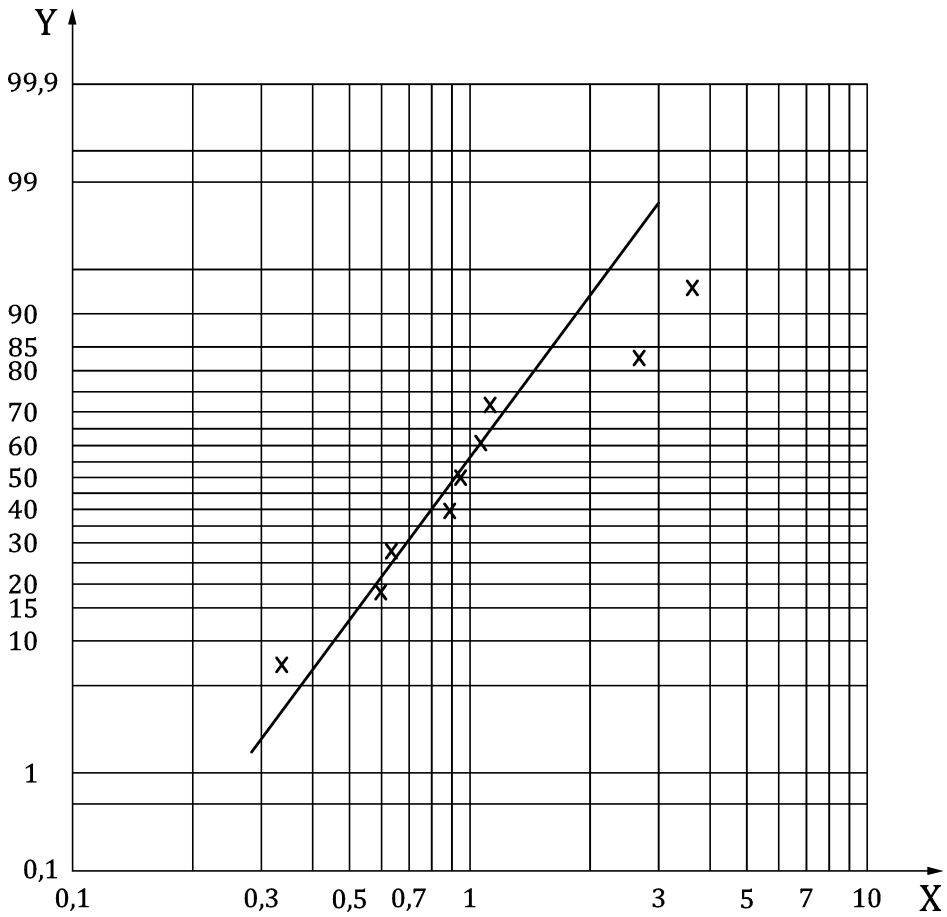
Regulations which require exposure not to exceed OELVs apply to every worker, and individuals who have consistently high exposure should be identified and not have their exposure disguised by incorrectly including them in a SEG with other workers.



Key

X exposure x_k (mg/m³)

Y probability P_k × 100

Figure E.4— Two of the exposure measurements in this SEG do not fit in with the rest of the distribution and may indicate exceptionally-exposed individuals

Of course, if individuals have exceptionally low exposures, then they will comply if the rest of the SEG does. However, their inclusion in the SEG can distort the compliance calculation for the SEG, so they should be removed from the SEG. It can be necessary then to make further measurements on the remaining SEG members to make up the minimum number required before the statistical test of compliance is made.

E.2.4.4 Distributions appearing to be not lognormal.

The exposure measurements plotted on Figure E.5 appear not to lie on a straight line, i.e not to be log-normally distributed; nor do they appear to lie on two intersecting lines, as in Figure E.2. This outcome is rare. The workers involved should be reconsidered to see if the SEG definition in 3.1.3 is met.

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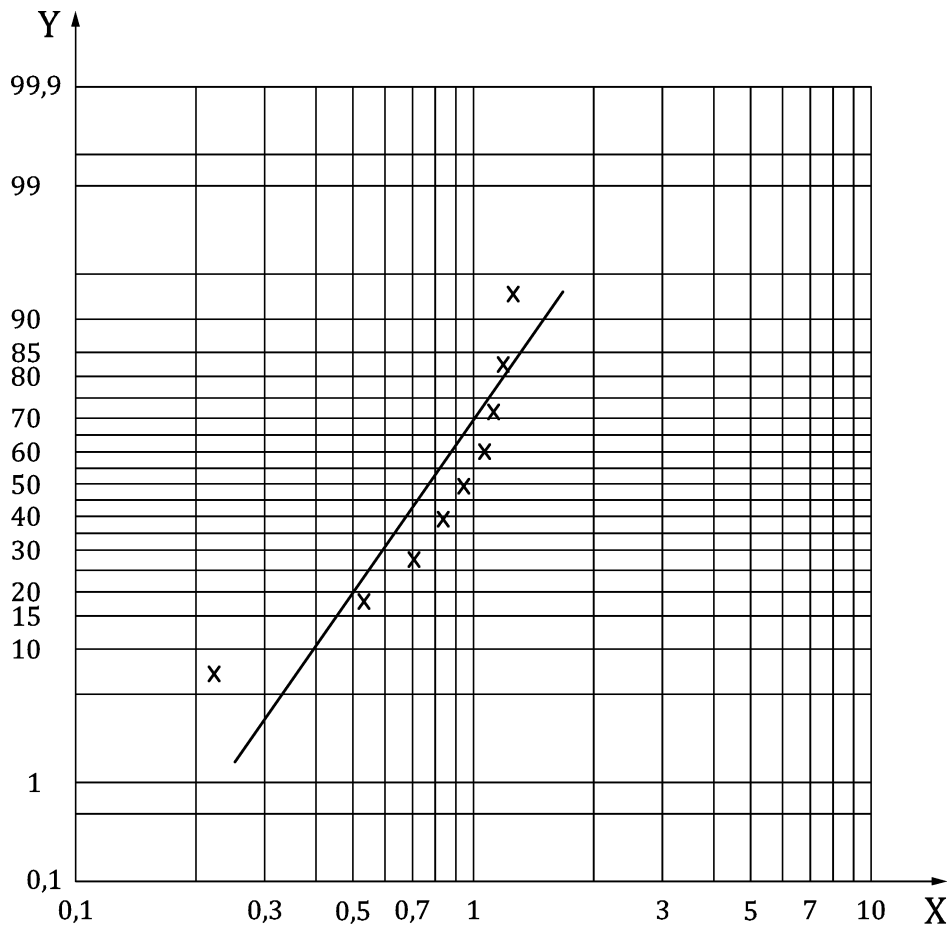

Key

X exposure x_k (mg/m³)

Y probability P_k × 100

Figure E.5 — Exposure measurements which appear not to be log-normally distributed

E.3 Statistical methods for the validation of SEGs

More rigorous statistical tests of the fit of the lognormal and other distributions to exposure results are included in data-handling software (for example, Altrex Chimie, BWStat, IHDataAnalyst, IHStat) but the power of such tests to identify non-lognormality is limited for the small sample numbers considered here. For example, of the cases presented in E.2, only the data in Figure E.5 is identified by the Shapiro and Wilk test[3] as inconsistent with a lognormal distribution.

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Annex F
(informative)

Testing compliance with OELVs

F.1 General

The approach recommended by this European Standard requires the implementation of a sampling strategy to test compliance using a limited number of measurements carried out in conditions representative of normal activities of a group of workers belonging to a SEG. A probability can then be calculated that all exposures of workers in the SEG comply with the OELV, but although this probability can be high it can never be 100 % because of the unavoidable variability of exposure.

A preliminary test using up to five measurements is given in 5.5.2. If this does not provide a decision, a statistical test meeting the specification stated in 5.5.3 should be used. The test given in this Annex uses at least six measurements, which may include the measurements made for the preliminary test. However, the more measurements that are made the more reliable is the diagnosis of compliance or non-compliance, provided that the measurements are representative of exposures in the SEG.

F.2 SEG compliance test for at least six exposure measurements

This test is based on the comparison of the 70 % upper confidence limit (UCL) with the 95th percentile of the distribution of the results [23]. If the UCL is greater than the OELV, it is concluded that there is an unacceptable probability of exceedance: the decision is **non-compliance**. If the UCL is less than the OELV, it is concluded that the probability of exceeding the OELV is acceptably low: the decision is **compliance**.

This test gives a balance between probability of declaring non-compliant a distribution which should prove compliant if a large number of samples were taken, and declaring compliant a distribution which should prove non-compliant if there were a large number of samples [7], [8], [9].

To perform this test and improve the reliability of diagnosis of compliance with the OELV, at least six measurements of exposure distributed amongst the workers in a SEG should be made. These should be representative of the exposure of the workers in the SEG, and this means that it can be necessary to make the measurements at different times from different days. This test could be used for log-normal and/or normal distribution of the results.

Log-normal distribution

If the distribution of results found to be log-normal (see Annex E), the geometric mean (GM) and the geometric standard deviation (GSD) of the set of results (x_i) should be calculated according to Formulas (F.1) and (F.2):

$$\ln(\mathrm{GM}) = \frac{\sum_1^n \ln(x_i)}{n} \qquad \mathrm{GM} = e^{\frac{\sum_1^n \ln(x_i)}{n}} \tag{F.1}$$

$$\ln(\mathrm{GSD}) = \sqrt{\frac{\sum_1^n \left(ln(x_i) - \ln(\mathrm{GM})\right)^2}{n-1}} \qquad \mathrm{GSD} = \mathrm{e}\sqrt{\frac{\sum_1^n \left(\ln(x_i) - \ln(GM)\right)^2}{n-1}} \tag{F.2}$$

The U_R variable calculated from the parameters set of results according to Formula (F.3) is compared with the tabulated value of U_T as a function of the number of results (see Table F.1).

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$$U_R = \frac{\ln(\mathrm{OELV}) - \ln(\mathrm{GM})}{\ln(\mathrm{GSD})} \tag{F.3}$$

Normal distribution

If the distribution of results found to be normal (see Annex E), the Arithmetic Mean (AM) and the standard deviation (SD) of the measurements should be calculated according to the Formulas (F.4) and (F.5):

$$AM = \frac{\sum_1^n x_i}{n} \tag{F.4}$$

$$SD = \sqrt{\frac{\sum_1^n (x_i - M)^2}{n-1}} \tag{F.5}$$

The variable U_R is in this case defined by Formula (F.6):

$$U_R = \frac{\mathrm{OELV} - AM}{SD} \tag{F.6}$$

F.3 Decision

If U_R is greater than or equal to U_T, then the conclusion is **compliance** with the OELV.

If U_R is smaller than U_T, then the conclusion is **non-compliance** with the OELV.

Table F.1 — U_T values depending on the number of results of exposure measurements

Number of exposure measurements *n*	U_T	**Number of exposure measurements** *n*	U_T	**Number of exposure measure ments** *n*	U_T
6	2,187	15	1,917	24	1,846
7	2,120	16	1,905	25	1,841
8	2,072	17	1,895	26	1,836
9	2,035	18	1,886	27	1,832
10	2,005	19	1,878	28	1,828
11	1,981	20	1,870	29	1,824
12	1,961	21	1,863	30	1,820
13	1,944	22	1,857	-	
14	1,929	23	1,851		

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Example F.1

A series of six exposure measurements following a log-normal distribution is used to test compliance with an OELV of 10 mg/m³.

Result in mg/m³	ln (result)
0,8	−0,223143551
0,9	−0,105360516
1,1	0,09531018
1,4	0,336472237
4,5	1,504077397
6	1,791759469
ln(GM)	0,566519203
ln(GSD)	0,863733553

GM = 1,76 mg/m³ ; *GSD* = 2,37.

Calculation of U_R

$$U_R = \frac{\ln(10) - 0,566519203}{0,863733553} = 2,009$$

The U_R value is lower than the U_T value (2,187) corresponding to six exposure measurements, concluding that there is still a probability that the OELV is exceeded: **Non compliance**.

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Annex G
(informative)

Exposure calculation with a work shift longer than 8h

Annex G applies to OELVs with a reference period of 8 h. Nowadays, many workers do irregular work schedule with work-shift longer than 8 h a day. For the exposure assessment of these workers, the daily exposure (E_d), given by Formula G.1, may be used to test compliance with OELVs.

$$E_d = C_i \times \frac{t}{8} \tag{G.1}$$

where

C_i is the work-shift occupational exposure concentration;

t is the associated exposure time, in hours.

By this approach, it is intended to give the same protection to extended work shift workers that is given to usual work shift ones. Other approaches can be found in the literature [9].

The following examples are given to illustrate how to use this approach.

Example G.1

There is a plant where workers start working at 8:30 h and finish at 19 h, having a dust exposure as indicated in Table G.1.

Table G.1 — Results of exposure measurements

Period of time	Task	Exposure mg/m³	Time h
8:30 to 13:00	production	5,3	4,50
13:00 to 14:00	lunch	no exposure	1,00
14:00 to 19:00	production	5,7	5,00

The occupational exposure concentration C_i can be calculated using the following Formulas:

$$C_i = \frac{(5,3 \times 4,5) + (5,7 \times 5,00)}{9,50} = 5,51 \text{ mg/m}^3 \text{ and } t = 9,50 \text{ h}$$

$$E_d = 5,51 \times \frac{9,5}{8} = 6,54 \text{ mg/m}^3$$

Example G.2

In a factory a workplace concentration of 1 mg/m³ of respirable dust is determined with a daily exposure time of 9 h. In this case, the resulting exposure (E_d) will be used for testing compliance with the OELV:

$$E_d = 1 \times \frac{9}{8} = 1,1 \text{ mg/m}^3$$

where

C_i = 1 mg/m³ and t = 9 h

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Annex H
(informative)

Exposure below the limit of quantification

H.1 General

Annex H gives a procedure to handle exposures that are below the limit of quantification (LOQ) when the statistical test given in 5.5.3 is used.

Results below LOQ should be taken into consideration since they represent existing but non-measurable exposure levels. They should not be removed, ignored or replaced with a fixed value such as LOQ/2, as these methods are likely to overestimate mean (GM or AM) and underestimate standard deviation (GSD or SD), which can lead to a wrong compliance decision.

Free of charge software that automates the procedure in this Annex is listed in H.5. It works quicker and with a higher precision, reproducibility, and reliability than the manual method, and its use is recommended. However, this Annex presents a graphical procedure which illustrates the method and which only requires the use of widely-available statistical tables.

H.2 Principle

It is assumed that all the exposures, including those below LOQ, approximate either a normal or lognormal distribution (see Annex E). A line is fitted to the results above LOQ on this assumption, and used to estimate the geometric mean (GM) and geometric standard deviation (GSD) of the distribution of all the measurements. The detailed procedure is illustrated in H.3.

H.3 Example

In this illustration, it is assumed that all the exposure measurements, including those below LOQ approximate to a lognormal distribution (see Annex E). In this example, the LOQ is 0,75 mg/m^3. Measurements were made until seven measurements gave results above LOQ. In the course of this, three results were obtained which were less than 0,75 mg/m^3. The results are shown in Table H.1 arranged in rank order. Figure H.1 shows the results as a log-probability plot, and the closeness to a straight line for the values above LOQ illustrates that they are lognormally distributed. The three values below LOQ are plotted with a symbol “e” at the LOQ. In reality of course they should appear at unknown positions at lower values of exposure, and they cannot be used to estimate the best line through the other points. The plotted probabilities P are shown in column 2 of Table H.1. They are calculated from the rank position k using Formula (H.1) given in ISO 5479. However, these calculated values are not in accordance with the uncertainty requirements given in EN 482.

$$P = (k - 3/8)/(N + 1/4) \qquad \text{(H.1)}$$

where

N is the number of points plotted, in this case 10.

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Table H.1 — Plotting points for the 10 exposure measurements

Rank k	**Likely fraction of a normal distribution less than the rank** R P_k [on y-axis in Figure H.1]	**Distance from the median (in standard deviation units) which corresponds to the fraction** P_k Z	**Exposure** x_k mg/m³ [on x-axis in Figure H.1]	**ln (exposure)**
1	0,061	− 1,547	< 0,75	-
2	0,159	− 1,000	< 0,75	-
3	0,256	− 0,655	< 0,75	-
4	0,354	− 0,375	0,80	−0,223
5	0,451	− 0,123	0,92	−0,083
6	0,549	0,123	1,05	0,049
7	0,646	0,375	1,1	0,095
8	0,744	0,655	1,45	0,371
9	0,841	1,000	2,0	0,683
10	0,939	1,547	2,5	0,916

The regression is derived for the seven points with exposure above 0,75 mg/m³.

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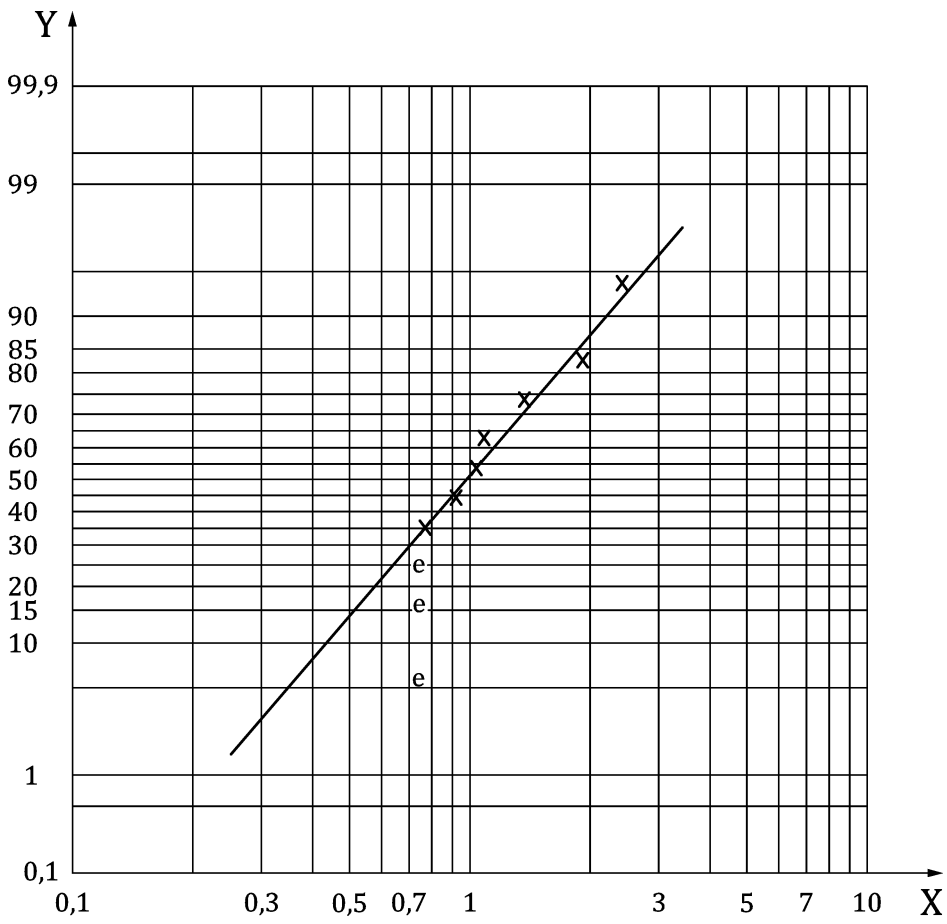

Key

X exposure x_k (mg/m³)

Y probability P_k × 100

e values below LOQ

Figure H.1 — Results plotted as a log-probability display

A regression line is drawn through the points above LOQ as a log-probability plot. The geometric mean (GM) and the geometric standard deviation (GSD) are obtained and can then be used in a statistical test as shown in Annex F.

Two methods are available, (a) the graphical method and (b) the calculation method:

a) Graphical method. The best fit line is drawn by eye through the results above LOQ. The Geometric Mean (GM) and Geometric Standard Deviation (GSD) are calculated from the line as follows. The Geometric Mean is the exposure corresponding to the fraction of the distribution at 0,5. The GSD is the ratio corresponding to the exposure fraction of 0,84 divided by GM.

b) Calculation method. The Z-score shown in the third column of Table H.1 is obtained from a spreadsheet, or if necessary from a table of the cumulative probability in the tail of a standard normal distribution, which can be found in any statistical text-book. If such a table is used, the fraction P (column 2 of Table H.1) is found in the body of the table, and the corresponding value of Z is read off from the row and column headings. This gives the plotting positions on the Z-axis of Figure H.2 corresponding to the exposures in column 4 of Table H.1.

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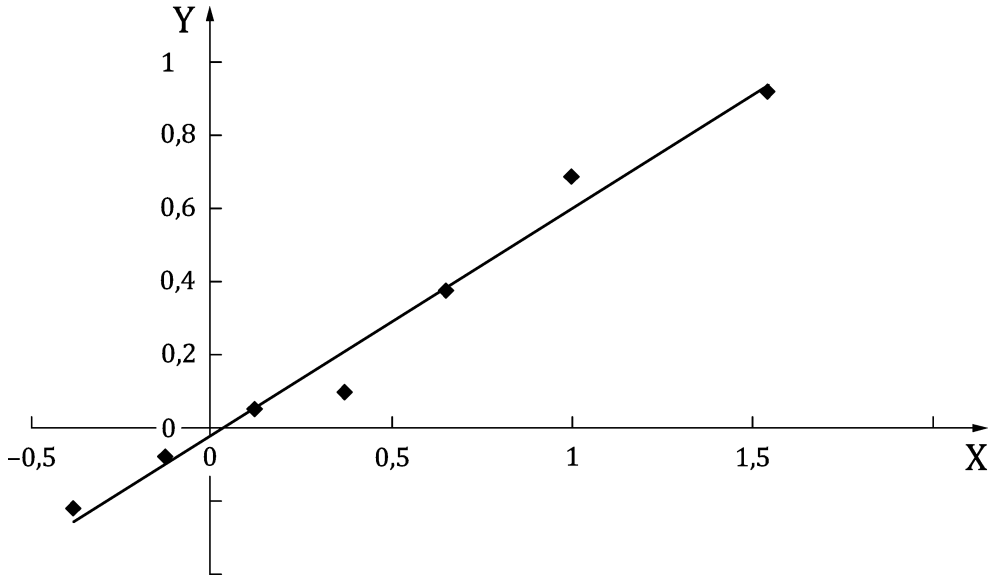
The natural log (ln) of the exposure (because this is a log-normal distribution) is then regressed on the Z-score for the seven points above LOQ, using a least-squares linear regression technique. This is explained in any statistics textbook [11] and is implemented in many calculators and statistics packages. The result is shown in Figure H.2. The fitted regression line in this case is:

$$\ln(\text{exposure}) = 0{,}619\ Z - 0{,}0248 \tag{H.2}$$

The regression coefficient (in this case 0,619) is the natural log of the GSD of the distribution, so GSD = exp(0,619) = 1,86 approximately, and the intercept −0,0248 is the natural log of the geometric mean exposure: GM = exp(−0,0248) = 0,976 mg/m³. The calculation method is more accurate than the graphical method, but the graphical method can easily be used as a rough check of the calculation.



Key

Y ln (exposure)

X distance from the median (in standard deviation units) which corresponds to the fraction P_k

Figure H.2 — Regression of ln (exposure) on Z for exposures above LOQ

To summarize, the regression produces ln(exposure) = *a*Z+*b, w*here *a* = ln(GSD) and *b* = ln(GM).

The above example applies to a lognormal distribution. If the distribution is normal, then the same procedure is followed, but Figure H.1 should use a normal probability display, and in Figure H.2 the exposure instead of ln (exposure) should be regressed on Z. Then in the graphical method, the SD is the difference between the exposures corresponding to P_k = 0,84 and P_k = 0,5 on Figure H.1 and the AM is the exposure corresponding to P_k = 0,5.

In the calculation method, if the regression is exposure = aZ+b, then a = SD and b = AM.

H.4 Uncertainty

The scatter of results about the regression line means that there is uncertainty in the regression coefficient (= ln(GSD)) and the intercept (= ln(GM)). In marginal cases this can affect the decision of whether the distribution complies with the OELV or not. Methods for quantifying the uncertainty of the regression are given in statistical texts [11].

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H.5 Software

Free of charge software that process nondetects are BWStat [12], HYGINIST [13], Altrex [14], NDExpo [16], IH Data Analyst (IHDA [27]). The estimates of the GM and GSD can be applied in the formulas of Annex F to test compliance. These tools can perform compliance tests as well.

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Annex I
(informative)

Interval for periodic measurements

After adequate prevention and protection have been achieved, reassessment of the situations should be conducted on a regular basis every year in order to assess if the exposure profile remains stable over time. Different approaches could be used to reach this goal (see Annex A). In parallel, some reassessments should include exposure measurements. Annex I gives three possible approaches to decide on the intervals between campaigns of periodic measurements. National regulations and other expert groups have suggested other intervals ▌AC⟩ [36, 37] ⟨AC▌.

The reassessment should be applied to each SEG, including an assessment of whether the allocation of the workers to the SEG is still appropriate.

a) If a compliance decision was reached by the preliminary test in 5.5.2 and conditions are unchanged, more measurements should be made to total at least six including the original preliminary tests, and examples b) and c) should be applied.

b) If more measurements per SEG were made, using for example the procedure in Annex F, then the geometric mean (GM) or the arithmetic mean (AM) of the earlier measurements, following respectively a log-normal or normal distribution can be used to decide the interval, as follows. After this interval the preliminary test given in 5.5.1 should be repeated and, if appropriate, the statistical test.

(GM or AM) < 0,1 OELV	36 months
0,1 OELV < (GM or AM) < 0,25 OELV	24 months
0,25 OELV < (GM or AM) < 0,5 OELV	18 months
0,5 OELV < (GM or AM)	12 months

c) If at least six measurements were made for a SEG in the initial compliance test, and the statistical test in Annex F was applied, the results may be used to decide the interval from the initial compliance test before periodic measurements, by testing whether the results indicates that the SEG complies with a fraction of the OELV as follows. Adapting Formula (F.3) to Formula (I.1 and I.2).

Log-normal distribution of the results (I.1)

$$U_R = \frac{\ln(j \times \mathrm{OELV}) - \ln(\mathrm{GM})}{\ln(\mathrm{GSD})} \qquad \text{(I.1)}$$

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Normal distribution of the results (I2)

$$U_R = \frac{j \times \mathrm{OELV} - \mathrm{AM}}{\mathrm{SD}} \qquad \text{(I.2)}$$

where

j	is a fraction as given below.
U_R	is compared with UT in Table F.1 to determine compliance with the fraction j of the OELV, as in Annex F. If the distribution is found to comply ($U_R > U_T$), then the interval before periodic measurements is given by:
$J < 0{,}25$	36 months
$0{,}25 < j < 0{,}5$	30 months
$0{,}5 < j < 1$	24 months

Example I1

Eight exposure measurements of toluene (OELV – 8 h is 192 mg/m³) were made in a SEG, with the following results in mg/m³:

Exposure measurements of toluene	ln(GM)	3,4982
29,5	ln(GSD)	0,4995
25,9	ln(OELV)	5,2575
28,0		
35,6		
104,8		
21,0		
35,3		
24,1		

Considering a log-normal distribution of results, compliance with the OEL is confirmed in accordance with the statistical test described in Annex F. The result of this test corresponds to a U_R value of 3,521 higher than the U_T value of 2,072 (see Table F.1) required to demonstrate compliance with OELV for eight exposure measurements.

Considering this U_T value of 2,072, the next step is to calculate the OELV fraction which is probably in compliance using the following Formula:

$$J = e^{(\mathrm{U_T} \times \ln(GSD) + \ln(GM) - \ln(\mathrm{OELV}))}$$

$$J = e^{((2{,}072 \times 0{,}4995) + 3{,}4982 - 5{,}2575)}$$

$$J = 0{,}484$$

According to the decision grid the next measurements should be conducted in 30 months, under the hypothesis that no major modifications of the exposure pattern occurs during this period.

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Waarom betaalt u voor een norm?
Normen zijn afspraken voor en door de markt, zo ook deze norm. NEN begeleidt het gehele normalisatieproces. Van het bijeenbrengen van partijen, het maken en vastleggen van de afspraken en het bieden van hulp bij de toepassing van de normen. Om deze diensten te kunnen bekostigen betalen alle belanghebbende partijen die aan tafel zitten voor het normalisatieproces, en u als gebruiker voor normen en trainingen. NEN is een stichting en heeft geen winstoogmerk.

Wat is nu precies de toegevoegde waarde van normen?
Stelt u zich eens voor … u wilt in het buitenland geld pinnen, maar uw bankpas past niet. Of uw nieuwe telefoon herkent uw simkaart niet. De samenstelling van de benzine over de grens is anders waardoor u niet kunt tanken. Het dagelijks leven zou zonder goede afspraken over producten, processen en diensten een stuk complexer zijn.

Het maken en vastleggen van afspraken door belanghebbende partijen noemen we het normalisatieproces. Normalisatie had vanouds betrekking op techniek en producten. Nu worden steeds vaker normen voor diensten ontwikkeld. Zo zijn er afspraken op het gebied van gezondheidszorg, schuldhulpverlening, kennisintensieve dienstverlening, externe veiligheid en MVO.

Normen zorgen voor verbetering van producten, diensten en processen; qua veiligheid, gezondheid, efficiëntie, kwaliteit en duurzaamheid. Dit ziet u op de werkvloer, in de omgang met elkaar en in de samenleving als geheel. Organisaties die normalisatie onderdeel van hun strategie maken, vergroten hun professionaliteit, betrouwbaarheid en concurrentiekracht.

Wat doet NEN?
NEN ondersteunt in Nederland het normalisatieproces. Als een partij zich tot NEN richt met de vraag om een afspraak tot stand te brengen, gaan wij aan de slag. We onderzoeken in hoeverre normalisatie mogelijk is en er interesse voor bestaat. Wij nodigen vervolgens alle belanghebbende partijen uit om deel te nemen. Een breed draagvlak is een randvoorwaarde. De afspraken komen op basis van consensus tot stand en worden vastgelegd in een document. Dit is meestal een norm. Afspraken die in een NEN-norm zijn vastgelegd mogen niet conflicteren met andere geldige NEN-normen. NEN-normen vormen samen een coherent geheel. Een belanghebbende partij kan een producent, ondernemer, dienstverlener, gebruiker, maar ook de overheid of een consumenten- of onderzoeksorganisatie zijn. De vraag is niet altijd om een norm te ontwikkelen. Vanuit de overheid komt regelmatig het verzoek om te onderzoeken of er binnen een bepaalde sector of op een bepaald terrein normalisatie mogelijk is. NEN doet dan onderzoek en start afhankelijk van de uitkomsten een project. Deelname staat open voor alle belanghebbende partijen. NEN beheert ruim 30.000 normen. Dit zijn de in Nederland aanvaarde internationale (ISO, IEC), Europese (EN) en nationale normen (NEN). In totaal zijn er ruim 800 normcommissies actief met in totaal bijna 5.000 normcommissieleden. Een goed beheer van de omvangrijke normencollectie en de afstemming tussen nationale, Europese en internationale normcommissies vereisen dan ook een zeer goede infrastructuur.

Betalen kleine organisaties net zoveel als grote organisaties?
Het uitgangspunt is dat alle partijen die deelnemen aan het normalisatieproces een evenredig deel betalen. De normcommissieleden kunnen onderling andere afspraken maken. Zo worden er wel eens afspraken gemaakt dat de grote partijen een groter deel betalen dan de kleinere bedrijven. De prijzen voor normen zijn voor iedereen gelijk. De kosten voor licenties zijn afhankelijk van de omvang van een organisatie en het aantal gebruikers.

Voordelen van normalisatie en normen
Gegarandeerde kwaliteit | Veiligheid geborgd | Bevordert duurzaamheid | Opschalen en vermarkten van nieuwe innovatieve producten | Meer (internationale) handelsmogelijkheden | Verhoogde effectiviteit en efficiëntie | Onderscheidend in de markt.

Voordelen van deelname
Invloed op de (internationale en Europese) afspraken | Als eerste op de hoogte van veranderingen | Netwerk; ook op Europees en internationaal niveau | Kennisvergroting.

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